

Rohit Suryaa Saravanan

AI / Machine Learning Engineer

Fulda, Hessen, Germany · +49 160 96597803 · Rohitsuryaas21@gmail.com
[linkedin.com/in/rohitsuryaa](https://www.linkedin.com/in/rohitsuryaa) · rohitsuryaa.com · github.com/srohitsuryaa21

Available from 15 July 2026 · Open to relocating



SUMMARY

AI/ML Engineer focused on taking machine-learning and LLM systems into production. MSc Data Science with two peer-reviewed IEEE publications in applied ML, plus a year of professional software engineering (REST APIs, SQL, CI/CD, Docker). I own the full path — data, modelling, evaluation, and deployment — and recently built an Agentic AI concierge with LangGraph, local LLM tooling, RAG, an eval harness, and pytest. English fluent (C1+); German A2 and actively improving.

EXPERIENCE

Associate Software Developer — EigenSecure (Remote, USA)

Mar 2024 – Apr 2025

- Built and shipped production backend services and RESTful APIs on an Agile team across MySQL, PostgreSQL, and MongoDB — the data and service layer that ML systems depend on.
- Optimised SQL queries, data workflows, and CI/CD pipelines for reliable deployments and traceable audit trails — the same automation and reproducibility discipline that MLOps requires.
- Translated ambiguous requirements into working systems and communicated trade-offs to both technical and product stakeholders.

SELECTED PROJECTS

Luxury Concierge Agent — Agentic AI

2026

- Built a local-first agentic AI assistant that turns client context (usage, region, budget, timeline) into guided recommendations.
- Engineered a LangGraph multi-step agent with local RAG over a curated knowledge base, tool-calling, conversation memory, a Typer evaluation CLI, and pytest checks.
- Stack:** Python, LangGraph, RAG, Ollama/qwen3.5:9b, ChromaDB, Streamlit, Typer, pytest · github.com/srohitsuryaa21/luxury-concierge-agent

Windmill Pitch Prediction — Applied ML (IEEE)

2024

- Reduced modelled forecasting error from 8.6% to 2.3% and lifted modelled energy output 21% by combining XGBoost with a residual-correction layer over six months of turbine data.
- Engineered 26 temporal features and validated via simulation and statistical evaluation; 96% prediction accuracy under varying conditions. Peer-reviewed IEEE publication.
- Stack:** Python, XGBoost, time-series feature engineering, simulation, statistical evaluation · DOI: [10.1109/ICEMCE57940.2023.10433983](https://doi.org/10.1109/ICEMCE57940.2023.10433983)

Supply Chain Intelligence — ML Forecasting Pipeline

2025

- Built an end-to-end ML pipeline over 110,123 transactions: SARIMA forecasting at sub-20% MAPE, EOQ inventory optimisation, and seller-risk scoring for 881 sellers.
- Delivered results to stakeholders through an interactive Plotly Dash decision dashboard.
- Stack:** Python, Pandas, NumPy, SARIMA, XGBoost, Plotly Dash, SQL · github.com/srohitsuryaa21/supply-chain-forecasting

Employee Attrition Prediction — Explainable ML

2023

- Built a classification pipeline on an HR dataset (EDA, feature engineering, model comparison) with explainable driver analysis so stakeholders could act on retention risk.
- Stack:** Python, Scikit-learn, Pandas, classification, explainability · github.com/srohitsuryaa21/employee-attrition-prediction

PUBLICATIONS

- Windmill Pitch Prediction with Adaptive Control** — IEEE ICEMCE 2023 — applied machine learning. DOI: [10.1109/ICEMCE57940.2023.10433983](https://doi.org/10.1109/ICEMCE57940.2023.10433983)
- DeFi Farmer Platform** — IEEE ICSCSS 2024 — blockchain-based agricultural finance. DOI: [10.1109/ICSCSS60660.2024.10625462](https://doi.org/10.1109/ICSCSS60660.2024.10625462)

EDUCATION

MSc Data Science — Hochschule Fulda, Germany

Apr 2025 – Present

Focus: Machine Learning, Data Mining & Predictive Analytics, Big Data & Distributed Systems, Process Mining, Research Methods, Data Ethics & GDPR.

BTech Big Data Computer Science — SRM University, Chennai, India

Aug 2020 – Nov 2024

Focus: Big Data systems, Distributed Systems, Algorithms, Databases, Software Engineering.

TECHNICAL SKILLS

Machine Learning: Python, PyTorch, TensorFlow, Scikit-learn, XGBoost, time-series, classification, model evaluation, explainability, statistical analysis

LLM & Agentic AI: LangGraph, Retrieval-Augmented Generation (RAG), local LLMs (Ollama), agent design, eval harnesses, ChromaDB, Streamlit

ML Engineering / MLOps: Docker, CI/CD (GitHub Actions, Jenkins), Azure, Git, REST APIs, pytest, Linux, YAML

Data: Complex SQL, Pandas, NumPy, ETL pipelines, feature engineering, PostgreSQL, MySQL, MongoDB

Software Engineering: C#, ASP.NET Core, Blazor, Java, Spring Boot (production software background)

CERTIFICATIONS

IBM Data Science Professional (Coursera)

LANGUAGES

English: Fluent (C1+) · German: A2 (actively improving) · Tamil: Native